



Protecting User Data Through Privacy-Sensitive Robot Design

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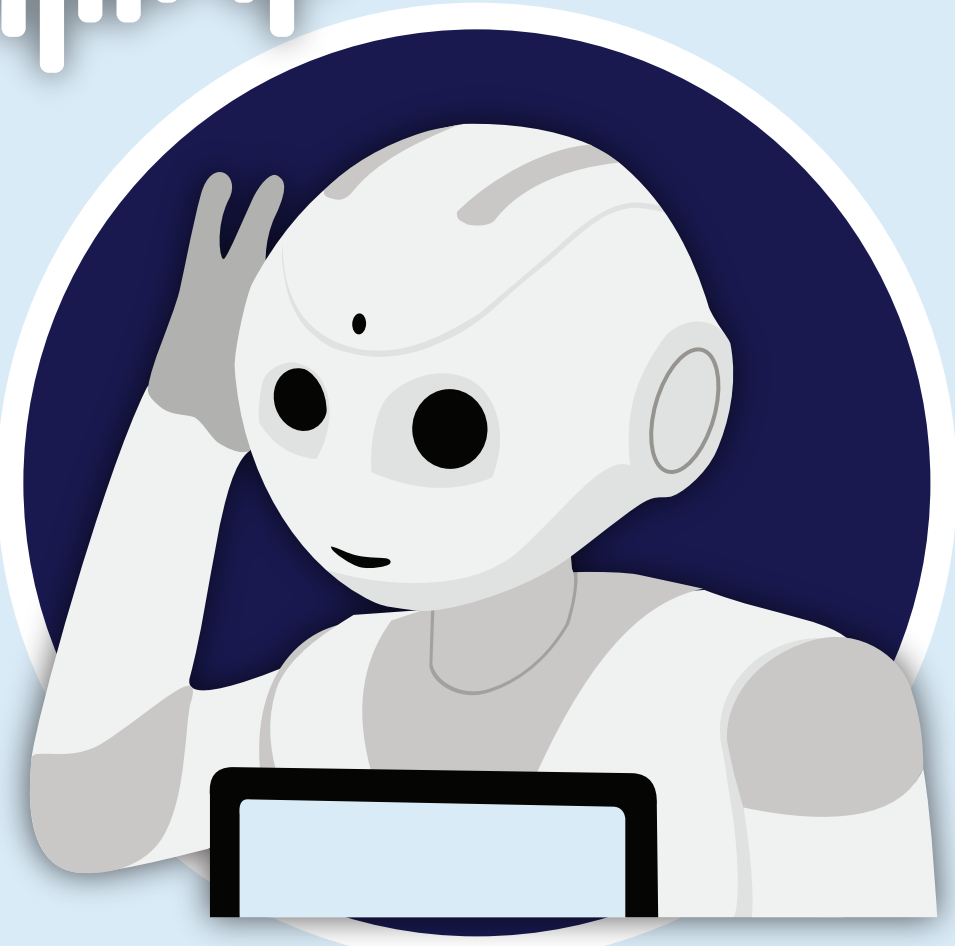
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Privacy-Sensitive Design in Human-Robot Interaction

Overview: Each of these works is intended to address privacy risks at the phases of data **collection**, **retention**, or **exposure**.

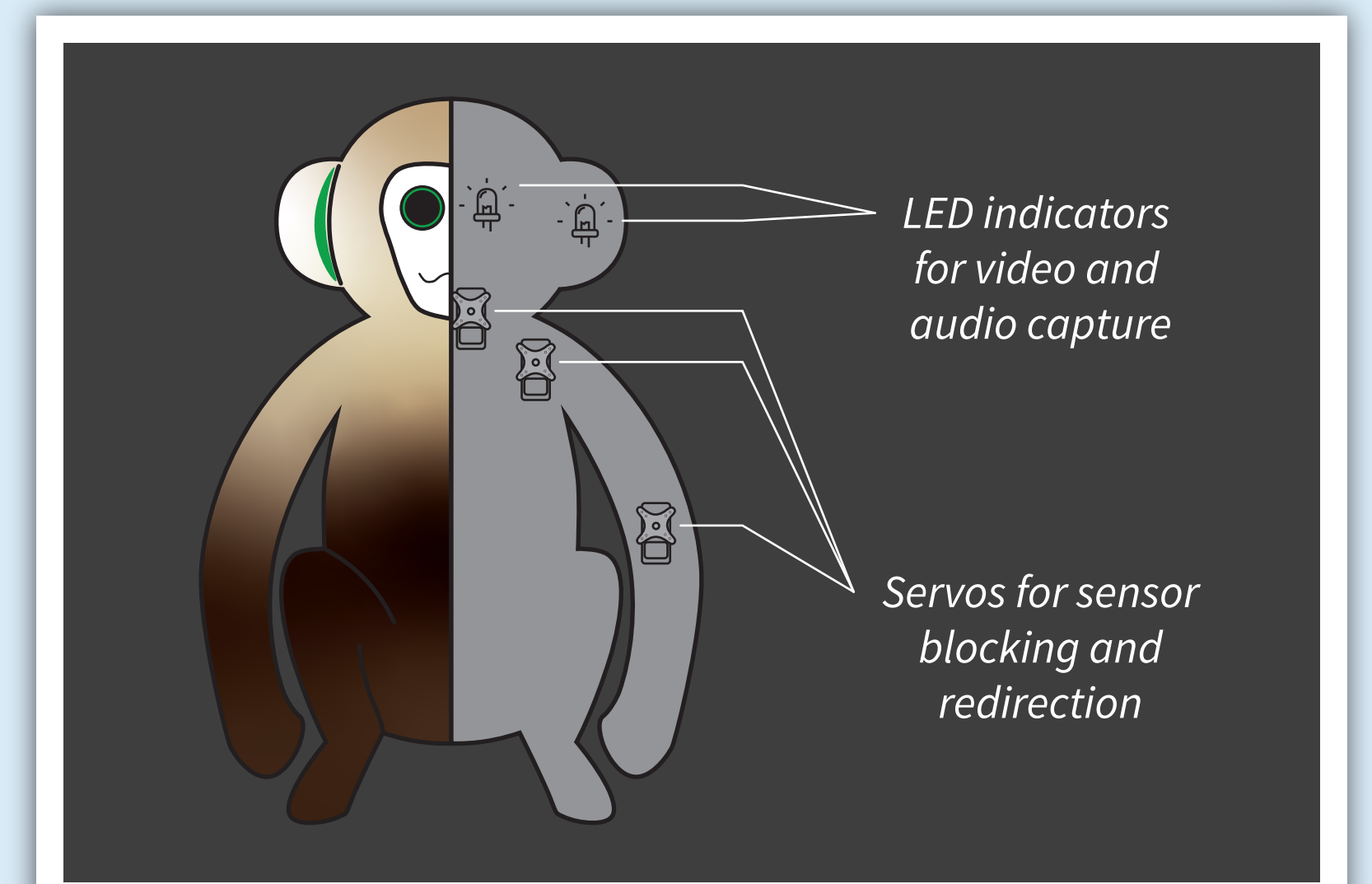
Goal: Our primary goal is to develop privacy-sensitive robot designs that **protect user data** and **mitigate risks** to everyday users with whom robots come into contact.

Data Collection



We are currently developing physical robot designs that enable a robot to **signal** and **block** its data collection. These designs are intended to influence **user awareness** of data collection and **inhibit sensitive data collection**.

Inhibiting Sensitive Data Collection



Data Retention

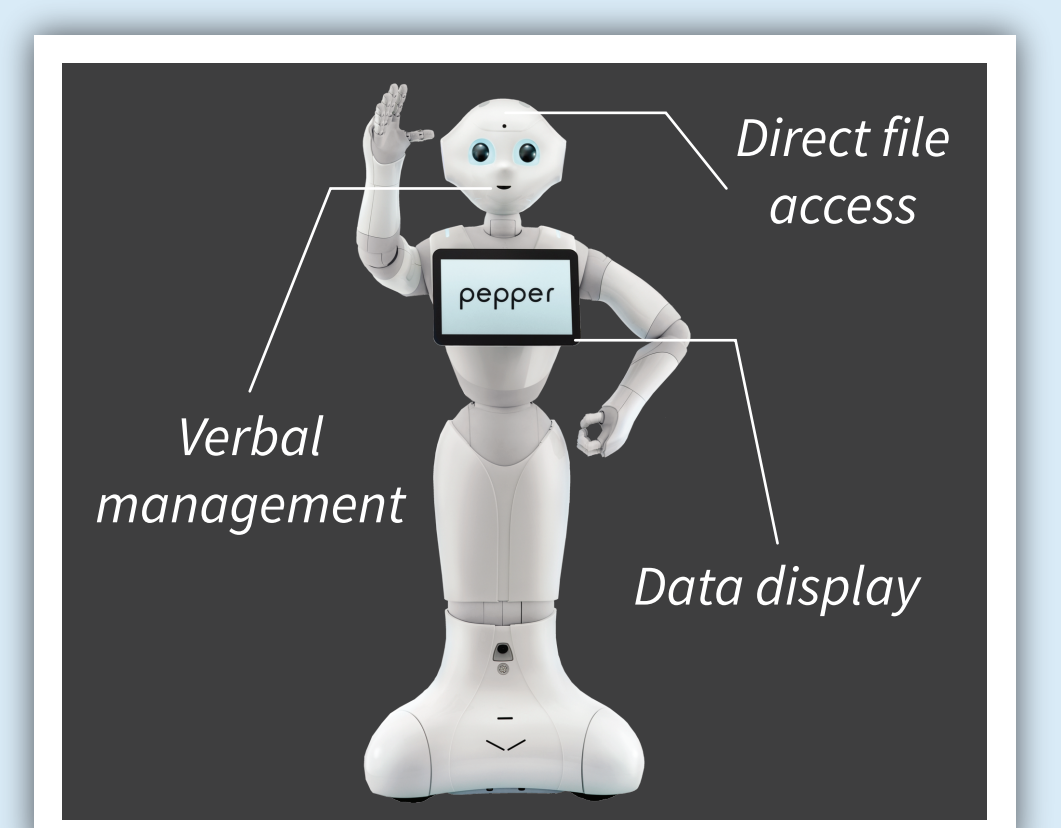


Exploring User Privacy Preferences

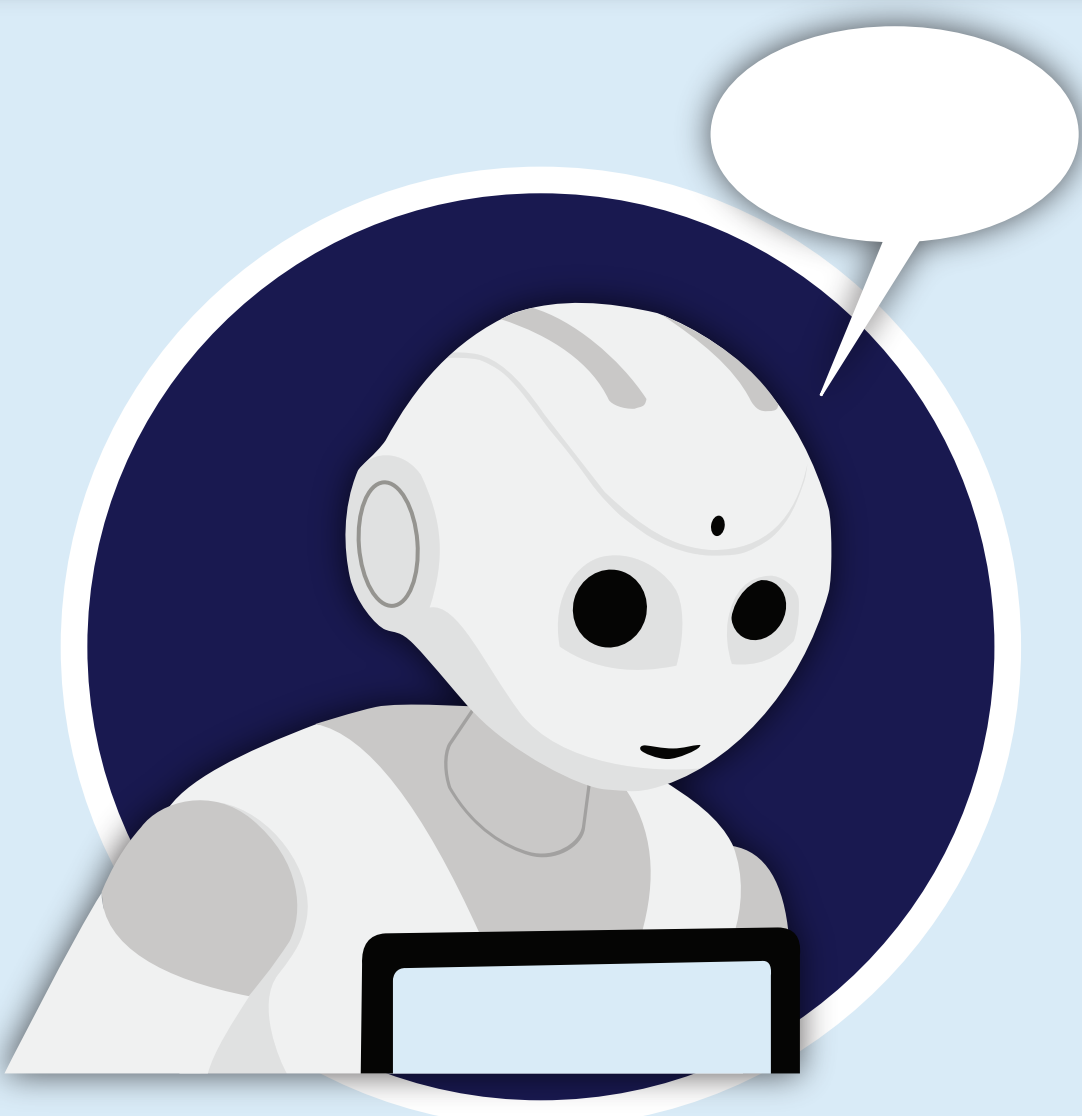
We intend to explore user **privacy preferences** in the presence and absence of robot **privacy threats** to illustrate the advantages and disadvantages of robot data collection.

Developing Data Management Prototypes

We will then leverage our findings to design, develop, and evaluate novel data management prototypes.

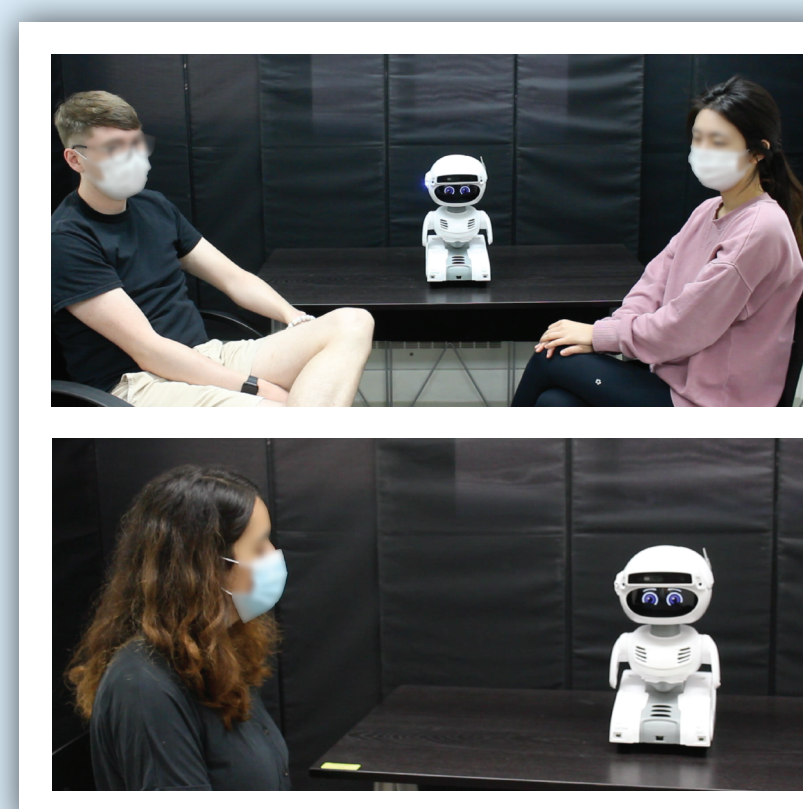


Data Exposure



Facilitating Appropriate Intentional Exposure

We developed a **privacy controller** for social robots that positively influenced perceptions of **privacy awareness**, **social awareness**, and **trustworthiness**.



Tang et al., HRI '22

Limiting Unintentional Exposure

We developed and evaluated three deceptive planning strategies intended to cause **confusion** or **misdirection** among hypothetical observers.

